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ABSTRACT

The transport process of two-phase fluids in porous media significantly affects groundwater seepage and carbon capture, utilization, and storage, as well as oilfield development. Due to limitations in analytical methods, the dynamic mechanisms controlling pore-scale flow processes remain insufficiently understood. Based on image analysis technology, this research elucidates flow instabilities triggered by the competition between capillary and viscous forces and clarifies their underlying mechanisms. Displaced-phase microcapillary number is introduced to provide a comprehensive analysis of the two-phase flow behavior. The results indicate that as Ca_{micro} increases, crude oil recovery factor declines, and the displacement process transforms from stable to unstable, with a critical Ca_{micro} of $10^{-4.09}$. When $Ca_{micro} = 10^{-4.33}$, viscous force dominates the flow process through Saffman–Taylor instability, exhibiting channeling with an extremely low recovery factor. When $Ca_{\mathrm{micro}} = 10^{-3.70}$, capillary force dominates the flow process. The rapid rebound of the local pressure causes a capillary burst, which makes the oil–water interface selection not follow the lowest threshold pressure rule. The sequential breakthrough effect, driven by the interaction between Saffman–Taylor instability and capillary burst, results in reduced residual oil in the swept area. As displacement progresses, the influence of viscous forces on oil–water distribution gradually weakens, while the impact of capillary forces becomes increasingly dominant. This transition is reflected in the weakening of the dynamic wetting effect and the formation of residual oil due to the Jamin effect.

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NOMENCLATURE

Symbols

Ca	Capillary numbers
Ca_{macro}	Macroscopic capillary number
Ca_{micro}	Microscopic capillary number
D	Etched pore depth
Eum	Modified Euler number
G_F	Growth factor
G_R	Growth rate
k	Microfluidic model permeability
L	Microfluidic model length
P_C	Capillary pressure
PX_{CO}	Amount of oil phase pixels in the current image
PX_{CR}	Amount of rock phase pixels in the current image
PX_{CW}	Amount of water phase pixels in the current image
PX_{IO}	Amount of oil phase pixels in the initial image
PX_{IR}	Amount of rock phase pixels in the initial image
PXP_i	Area covered by pores with a radius of i
PXR_i	Area covered by region with a number of i
Q	Injection rate
R	Average pore radius
r	Pore radius
RFP_i	Recovery factor of pores with a radius of i
RFR_i	Recovery factor of region with a number of i
S	Swept area in the model
t	Actual experimental time
T_{BT}	Breakthrough time of oil–water front
v	Linear velocity in a microfluidic model
W	Microfluidic model width
β_0	Number of objects in the target area
β_1	Number of holes within the target object
η_o	Oil viscosity
θ	Contact angle
σ	Oil–water interfacial tension
τ	Dimensionless time
Φ	Microfluidic model porosity

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Abbreviations

EOR	Enhanced oil recovery
FD	Fractal Dimension
IFT	Interfacial tension
RF	Recovery factor
RPM	Revolutions per minute
TDS	Total dissolved solids

I. INTRODUCTION

Two-phase flow in microchannels occurs in many natural and industrial processes, such as enhanced oil recovery (EOR),[1–3] carbon dioxide storage,[4,5] groundwater seepage,[6,7] drug delivery in blood vessels,[8] food industry,[9] and nanoparticle synthesis.[10] A representative example is immiscible displacement, a laminar multiphase flow governed mainly by interfacial tension (IFT), capillary forces, and viscosity.[11–13] Under certain combinations of pore geometry, fluid viscosity, and injection rate, oil–water flow becomes highly unstable, reducing the recovery factor.[14,15] Therefore, studying the characteristics of the oil–water two-phase displacement process is of great significance to the petroleum industry.[16,17]

The complex macroscopic flow patterns result from the competition between microscale capillary forces, viscous forces, and other forces. Over the past century, researchers have investigated pore-scale phenomena through two primary routes: laboratory experiments[18,19] and numerical simulations.[20,21] Microfluidics, an interdisciplinary technique, offers real-time, objective, and reproducible visualization at the pore scale.[22,23] Laser etching has enabled the fabrication of porous-network chips that imitate geological pore structures.[24] This has greatly advanced the study of two-phase fluid behavior in complex underground pore spaces, attracting significant attention.[25–27] However, existing micromodels face a trade-off between resolution and field of view. High-resolution devices are typically small, hindering observation of global interface evolution.[28,29] Conversely, large-scale models, such as sand-pack models, make it difficult to analyze pore-scale details.[30,31] Recent technical progress has begun to mitigate these limitations. For instance, micro-nano X-ray CT and laser confocal microscopy capture three-dimensional pore networks and partly close the resolution–scale gap.[32–35] However, these methods involve high costs and long reconstruction times, which hinder real-time detection. The complexity of experimental parameters, observation methods, and processing techniques presents additional challenges for studying two-phase flow in porous media. Numerical models avoid physical constraints on size, structure, and fluid properties and can explore more complex scenarios.[36–38] However, their predictive accuracy often remains uncertain and must be validated against experiments. Therefore, large-scale microscopic visualization model analysis of oil–water two-phase flow at the pore level remains necessary.

Scholars have also conducted in-depth research on the characteristics of pore-scale two-phase flow, the underlying mechanical mechanisms, and the resulting macroscopic manifestations. Lenormand systematically established a displacement mode phase diagram for immiscible two-phase fluids based on flowability ratio and capillary number, dividing the displacement process into three modes: capillary fingering, viscous fingering, and stable displacement.[39] Subsequent studies quantified the effects of pore architecture, wettability, injection rate, and fluid properties using fractal dimension, saturation distribution, and related parameters.[40–44] On this basis, the transition zone between capillary fingering and viscous fingering has been identified.[12,45] In addition, the influence of frictional fluid dynamics in particle flow on two-phase flow patterns has been examined.[46] Al Housseiny demonstrated that convergent Hele–Shaw elements with negative gradients can suppress unstable flows.[47] Qiu highlighted the importance of roughness on two-phase flow and fluid distribution in the later stage.[48] Lu argued that the queuing effect caused by pressure differences is the fundamental reason for variations in two-phase flow in porous media.[49] These studies suggest that flow patterns result from the competition between two or more dynamic mechanisms. However, due to analytical limitations, existing pore-scale research still relies mainly on semi-quantitative inference, lacking detailed descriptions and direct quantitative evidence at the pore scale. Developing a set of pore-scale analytical methods is crucial for a deeper understanding of the mechanical mechanisms and macroscopic manifestations of two-phase flow.

A large-scale microfluidic chip was designed to capture macroscopic flow patterns while resolving pore-scale two-phase behavior. Image-processing algorithms were then implemented to quantify, in real time, displacement efficiency, flow paths, capillary pressure, and other key parameters. Together, the chip and algorithms elucidate the pore-scale mechanisms that govern complex two-phase flow phenomena.

II. EXPERIMENTAL METHOD AND MATERIALS

A. Experimental fluids

The oil used in this study is crude oil from four blocks of Liaohe Oil Field, named crude oil A, B, C, and D, respectively. The synthetic water has a TDS of 3384.0 mg/l, and the ion composition is shown in Table I. To emphasize the differences among rock particles, oil, and water, methyl blue was consistently used to stain the synthetic water.

B. Microfluidics setup and experimental procedure

As shown in Fig. 1(a), the microfluidic experimental setup consists of a displacement system and an acquisition system. The displacement system includes a Harvard PHD ULTRA Symphony pump to provide displacement energy, a SEENO LED surface light source to provide sufficient illumination, a waste fluid tank for collecting waste liquid, and several pipelines. The acquisition system comprises a Sony A7M4 camera with an FE90 mm F2.8 macro lens and a computer for data recording.

A glass microfluidic chip was used in this study [Fig. 1(b)], featuring a pore structure composed of repeating basic units derived from cast thin sections of real rock cores.[12,50,51] The chip fabrication process is described in detail by Zhao *et al.*.[52] The etched pore depth is 13 μm (the cross-section shape of the channel is a rectangle), and the chip measures 10 cm in length and 4 cm in width. The pore radii range from

TABLE I. Ionic composition of synthetic water.

Ion	K^++Na^+	Ca^{2+}	Mg^{2+}	Cl^-	SO_4^{2-}	HCO_3^-	CO_3^{2-}
Concentration (mg/l)	960.5	20.1	9.1	238.2	24.0	2051.4	106.3

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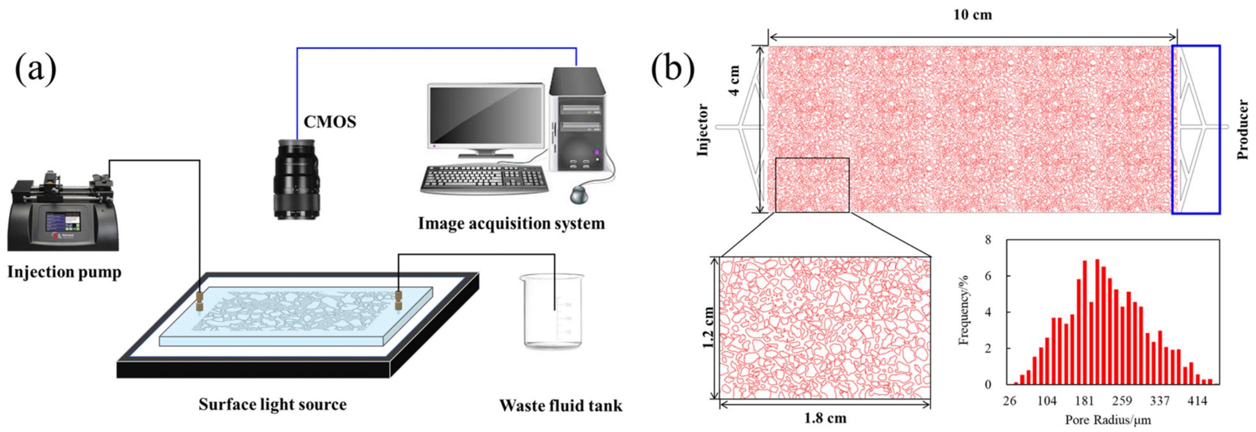

FIG. 1. The schematic diagram of micromodel flooding. (a) Experimental device assembly. (b) Chip design diagram and pore radius distribution.

26 to 453 μm, with an average of 225 μm. It is worth noting that the model used in this study has an overall size of $100\,000 \times 40\,000\,\mu m^2$. During imaging, the pixel resolution is 7676×3079 in the horizontal and vertical directions, respectively, resulting in a pixel size of 13.028 μm/pixel. This resolution corresponds to 1/4 of the minimum pore radius and 1/35 of the average pore radius, which is sufficient for accurate image-based analysis. The model design includes four branch injection channels connected to a horizontal channel at both the inlet and outlet ends [see blue box in Fig. 1(b)], ensuring uniform fluid distribution across the entire chip.

The microfluidic chip was first saturated with crude oil obtained from four different blocks of the Liaohe Oil Field. Subsequently, synthetic water was injected into the model at a constant flow rate of 1 μl/min. A camera recorded the displacement process at 30 s intervals. To ensure objectivity and reliability, each experiment was conducted over a sufficient duration until the oil–water distribution reached a steady state.

C. Fluid property tests

The IFT between crude oil and synthetic water was measured using a TX-500C IFT meter (CNG Corporation, USA) at a speed of 5000 RPM. The viscosity of crude oil was measured using a TC-150SD viscometer (Brookfield Corporation, USA). The static contact angle was measured using a Theta Flex optical tensiometer (Biolin Scientific, Sweden).

III. IMAGE PROCESS METHOD

A. *In situ* calculation method for capillary pressure

Capillary pressure refers to the pressure that causes a wetting or non-wetting liquid to spontaneously rise or fall within a capillary, reflecting the interaction between liquid–liquid and liquid–solid phases. In the calculation of capillary pressure [Eq. (1)], the σ can be measured using an IFT meter. So the key to calculating *in situ* capillary pressure is the measurement of contact angle and pore radius (Fig. 2).

$$P_c = \frac{2\sigma \cos\theta}{r}. \tag{1}$$

1. Image preprocessing

Accurate identification of rock (R), crude oil (O), and water (W) distributions is essential. A series of morphological operations was applied as detailed in the supplementary material, Sec. S1.

2. In situ measurement method for the three-phase contact angle

a. Extraction of phase interfaces and determination of search domain. To locate three-phase contact points (i.e., water–oil–rock intersections), pixel-wise evaluation is needed, which is computationally intensive. Since these points only occur at three-phase boundaries, the interface is first extracted to narrow the search area, significantly reducing processing time (supplementary material, Sec. S2).

b. Localization of the three-phase contact point based on contact relationship discrimination. After determining the search domain, precise localization of the three-phase contact point is required, which involves analyzing each pixel within the search domain. This step follows the definition method of the three-phase contact point detailed in the supplementary material, Sec. S3.

c. Contact angle calculation based on interface fitting. Contact angles are obtained by fitting two lines along the oil–water and oil–rock interfaces passing through the three-phase contact point. The angle between these lines represents the contact angle (supplementary material, Fig. SF-1).

3. In situ calculation method for capillary pressure

a. Oil–water interface tracking and contact point classification. Based on the oil–water interface characteristics, with the classification of rock, crude oil, and water phases already completed, morphological methods are used to determine the oil–water interface. This involves first converting the ternary image into a binary image of water and other components, then extracting the oil–water interface through erosion and subtraction operations on the binary image (supplementary material, Fig. SF-2). The endpoints are matched with previously

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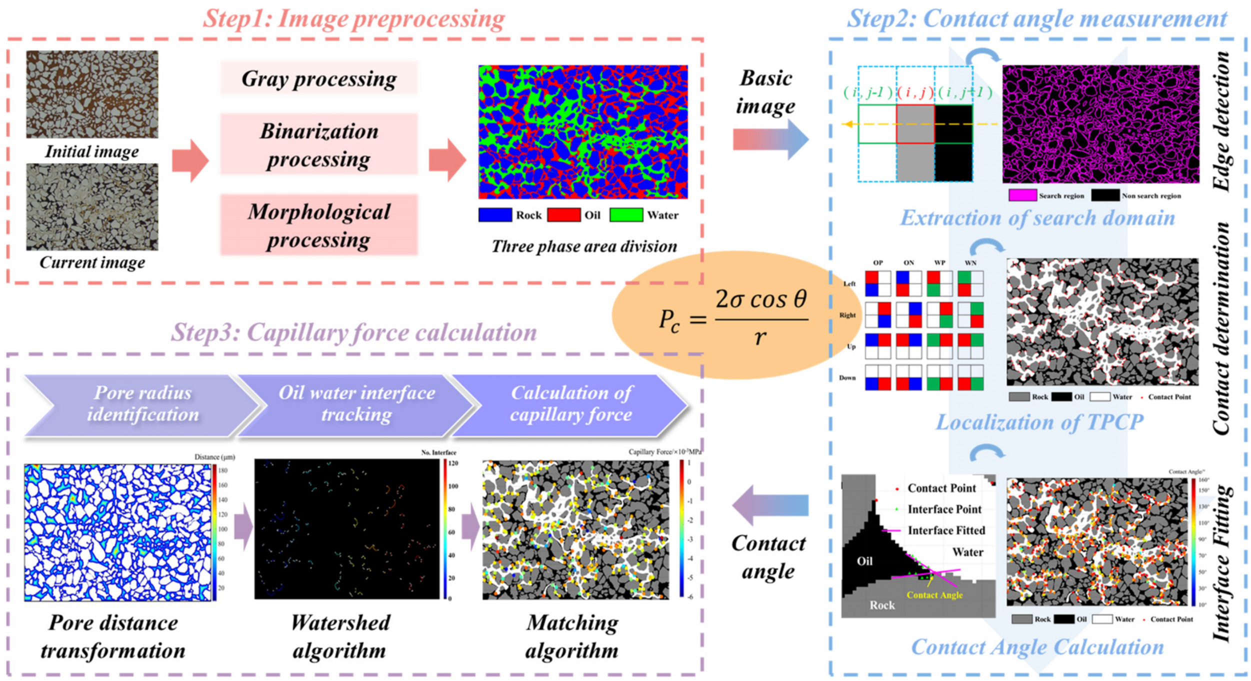

FIG. 2. Flowchart of *in situ* capillary pressure calculation.

extracted three-phase contact points based on the principle of minimum distance, as shown in the supplementary material, Fig. SF-3.

b. Pore radius identification and capillary pressure calculation. An axis-based method is employed to estimate the pore radius. Capillary pressure is then computed using Eq. (1) (supplementary material, Sec. S4).

B. Oil recovery factor calculation of microchip and pores with different sizes

The model's recovery factor was determined by analyzing changes in oil, water, and rock pixels [Eq. (2)]. Recovery factor for pores of different sizes was determined to be the same within regions of specific pore radii. Details are provided in the supplementary material, Sec. S5. Similarly, as shown in Eq. (4), the recovery factor across different regions (i.e., its variation with distance) was obtained by partitioning the image and calculating each region separately (supplementary material, Fig. SF-4)

$$RF = \frac{PX_{IO} - PX_{CO}}{PX_{IO}} \times 100\% = \frac{PX_{CW}}{PX_{IO}} \times 100\%, \tag{2}$$

$$RFP_i = \frac{PX_{IO} \cap PXP_i - PX_{CO} \cap PXP_i}{PX_{IO} \cap PXP_i} \times 100\% = \frac{PX_{CW} \cap PXP_i}{PX_{IO} \cap PXP_i} \times 100\%, \tag{3}$$

$$RFR_i = \frac{PX_{IO} \cap PXR_i - PX_{CO} \cap PXR_i}{PX_{IO} \cap PXR_i} \times 100\% = \frac{PX_{CW} \cap PXR_i}{PX_{IO} \cap PXR_i} \times 100\%. \tag{4}$$

C. Calculation method for swept path and interfacial property

In order to study the dynamic sweep patterns and the stability of the oil–water front under various conditions, the swept path and fractal dimensions of oil–water front were calculated separately. For swept path identification, a recognition method based on a sequential swept diagram, in which pixel values represent the time when the injected water affects the pixel, and the minimum path method was adopted. Specifically, the total swept area at different moments was determined through image binarization. Sequential subtraction of these images produced the current swept area, which was used to create a sweep sequence diagram, where each index represents the moment when a pixel was swept. After applying special processing to the rock particle and boundary areas of the image, Dijkstra's algorithm was used to calculate the minimum path from the inlet to the outlet side, identifying the swept path (Fig. 3). Dijkstra's algorithm is based on a greedy strategy, which gradually expands the known shortest path region to find the shortest path tree from the source point to all other nodes.

The fractal dimension can be used to characterize the stability of the displacement process, and the larger the fractal dimension, the more unstable the displacement process. The calculation of the fractal dimension for the oil–water front was relatively straightforward and is explained in the supplementary material, Sec. S6.

IV. RESULT AND DISCUSSION

A. Fluid property and verification of the capillary pressure calculation method

Figure 4(a) shows the measurement results of viscosity, interfacial tension (IFT), and static contact angle across the four crude

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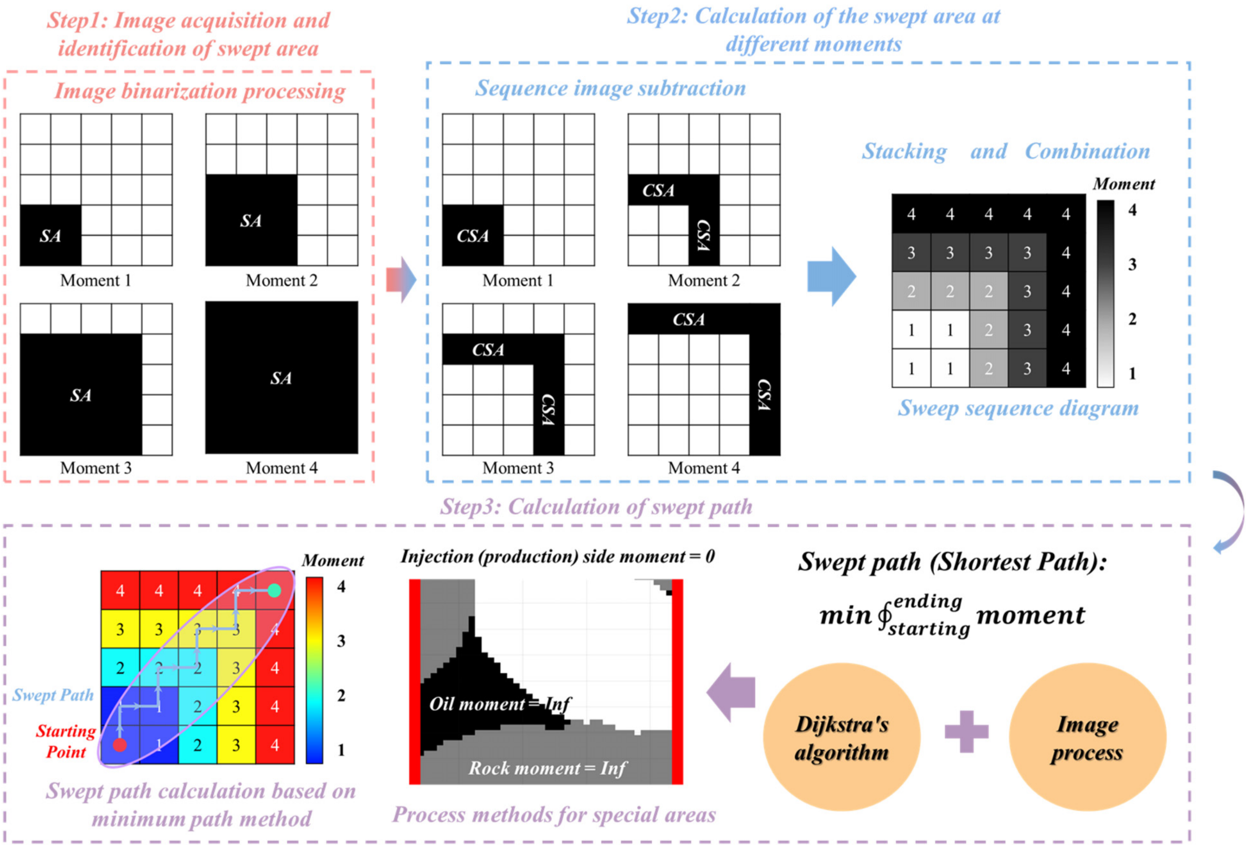

FIG. 3. Flowchart of swept path calculation. (SA: Swept Area, CSA: Current Swept Area).

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oil samples. Verification of the capillary pressure calculation required independent validation of the contact angle and pore radius measurements. First, the detected contact angles are validated using manual measurement as an auxiliary means. The same approach is applied to validate the pore radius measurements. As shown in Fig. 4(b), the contact angle and pore radius calculated by the method proposed in this article have good consistency with the measurement results.

B. Transition from stable to unstable displacement based on microscopic capillary number in porous media

Dimensionless time was introduced for the subsequent analysis in this research, which is defined as the ratio of actual time to breakthrough time [Eq. (5)]. Figure 5(a) reveals two stages in the micro-displacement process. In the displacement stage, water gradually advances to replace the oil in the pores. In the breakthrough and channeling stage, after the oil–water front reaches the outlet, injecting water channels along the early-formed channeling, and the recovery factor tends to stabilize. Overall recovery factor declines as crude oil viscosity increases. Crude oil A, despite having the lowest viscosity, retains some remaining oil in the swept area, causing strong curve volatility. Crude oil B exhibits a more uniform overall recovery, while crude oil D, with the highest viscosity, has low sweep efficiency due to severe channeling [Fig. 5(b)]

$$\tau = \frac{t}{T_{BT}}. \tag{5}$$

A power function was used to fit the relationship between the fractal dimension of the oil–water front and dimensionless time. During water flooding, the oil–water front evolves from a regular to an irregular geometry [Fig. 5(c)]. The power law coefficient ("growth factor") and exponent ("growth rate") quantify this geometric evolution. When viscosity exceeds 25 mPa s, both parameters rise sharply, indicating that highly viscous oils destabilize the front more quickly

$$FD = G_F \tau^{G_R}. \tag{6}$$

The microscopic capillary number Ca_{micro} is employed to characterize water–oil two-phase flow at the pore scale [Eq. (7)].[53] Because our experiments vary the viscosity of the displaced phase, this research adopts the displaced-phase Ca_{micro} throughout, which is defined as the ratio of viscous force and capillary force (also the ratio of viscous stress and capillary pressure) on the micromodel.[54] To highlight the flow characteristics within porous media, this study also introduces the macroscopic capillary number Ca_{macro} [Eq. (9)]

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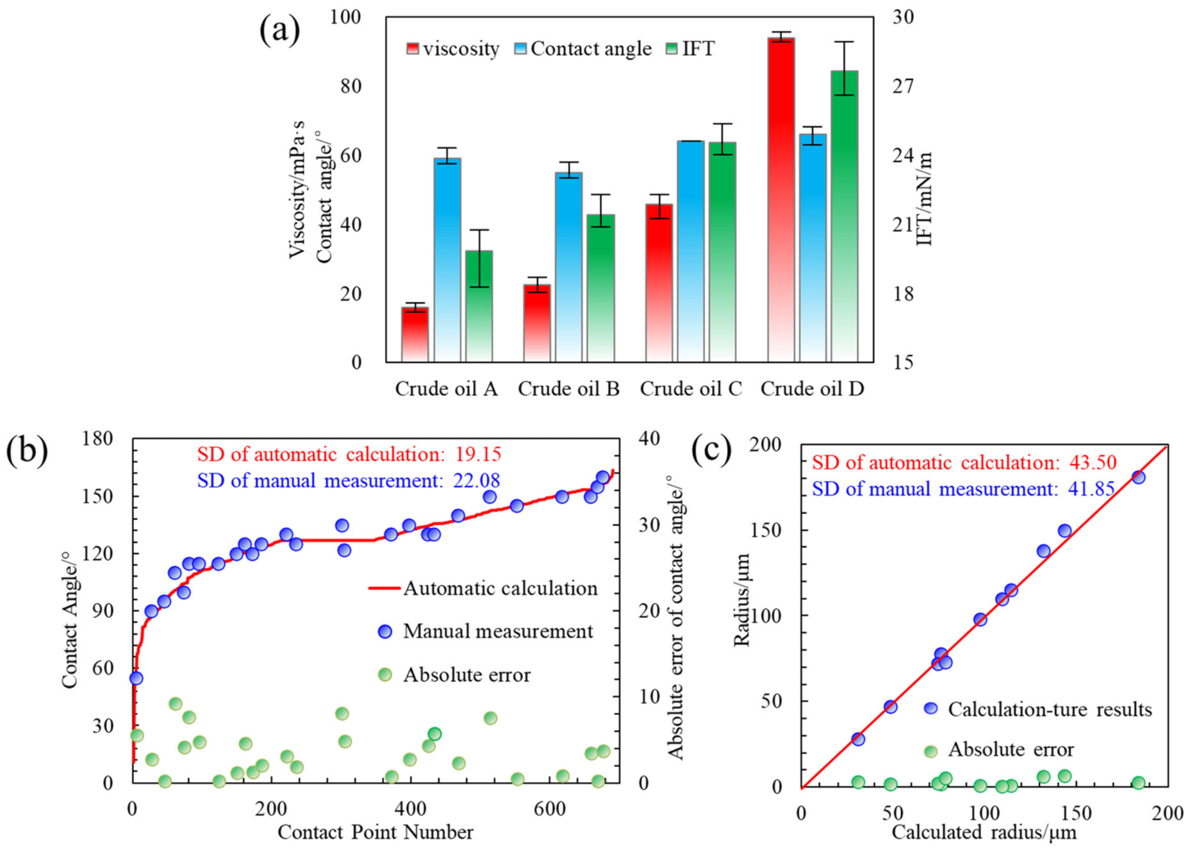

FIG. 4. Basic properties of crude oil and verification of the calculation method of capillary pressure. (a) Experimental fluid property measurement results. (b) Verification of automatic contact angle measurement results. (c) Verification of pore radius measurement results.

$$Ca_{\text{micro}} = \frac{\eta_o v}{\sigma}, \tag{7}$$

$$v = \frac{Q}{DW\Phi}, \tag{8}$$

$$Ca_{\text{macro}} = \frac{M\Phi vL}{kP_c} = Ca_{\text{micro}}\frac{\Phi LR}{2k}, \tag{9}$$

where v is the linear velocity in a micro model with pores depth $D = 13\,\mu$m, chip width $W = 40$ mm, porosity $\Phi = 55\%$ at the injection rate of Q. $L = 100$ mm is the length of micromodel, $R = 225\,\mu$m is the average pore radius, $k = 2.42 \times 10^{-9}$ m^2 is the permeability of micromodel, calculated by the Kozeny–Carman equation.[55]

Modified Euler number was introduced to illustrate the characteristics of displacement processes [Eq. (10)]. Given that the displacement media coverage varies under different conditions, the Euler number[56] per unit area is employed for characterization. It is not difficult to see that the smaller the Eum, the more dispersed the remaining oil in the affected area. Information about this part is detailed in the supplementary material, Sec. S7

$$Eum = \frac{\beta_0 - \beta_1}{S}. \tag{10}$$

Figure 6(a) demonstrates that swept area morphology directly reflects displacement stability. The data reveal a critical line that separates stable and unstable displacement modes according to Ca_{micro}. For $Ca_{micro} = 10^{-4.33}$,$10^{-4.21}$, the water flooding process can be considered stable, considering that they have a more stable oil–water front and more similar and smaller fractal dimensions [Fig. 5(c)]. For $Ca_{micro} = 10^{-3.96}$,$10^{-3.70}$, the flow process exhibits unstable characteristics, prone to fingering, significantly reducing displacement efficiency. The critical value of the logarithm of Ca_{micro} is -4.09. Stable flows increase the recovery factor of the model by 47.33%, and the front stability by 20.3%. Additionally, less remaining oil is present in the swept area, and Eum increases from -304 to -148 with increasing Ca_{micro}. When $Ca_{micro} = 10^{-4.33}$, a distinct lag separates breakthrough from final recovery; this breakthrough behavior typifies transitional flow [Figs. 6(b) and 6(d)].

C. Pore-scale flow characteristics of different Ca_{micro} controlled by viscous and capillary force

1. Pore-scale recovery factor characteristics

Figure 7 presents the pore-throat recovery factor for crude oils A–D. The injected water first invades larger pores and subsequently channels into smaller ones, so pores larger than 100 μm retain an almost constant recovery factor. Taking a recovery factor threshold of 25% as the criterion, the minimum effective pore-throat radii for oils

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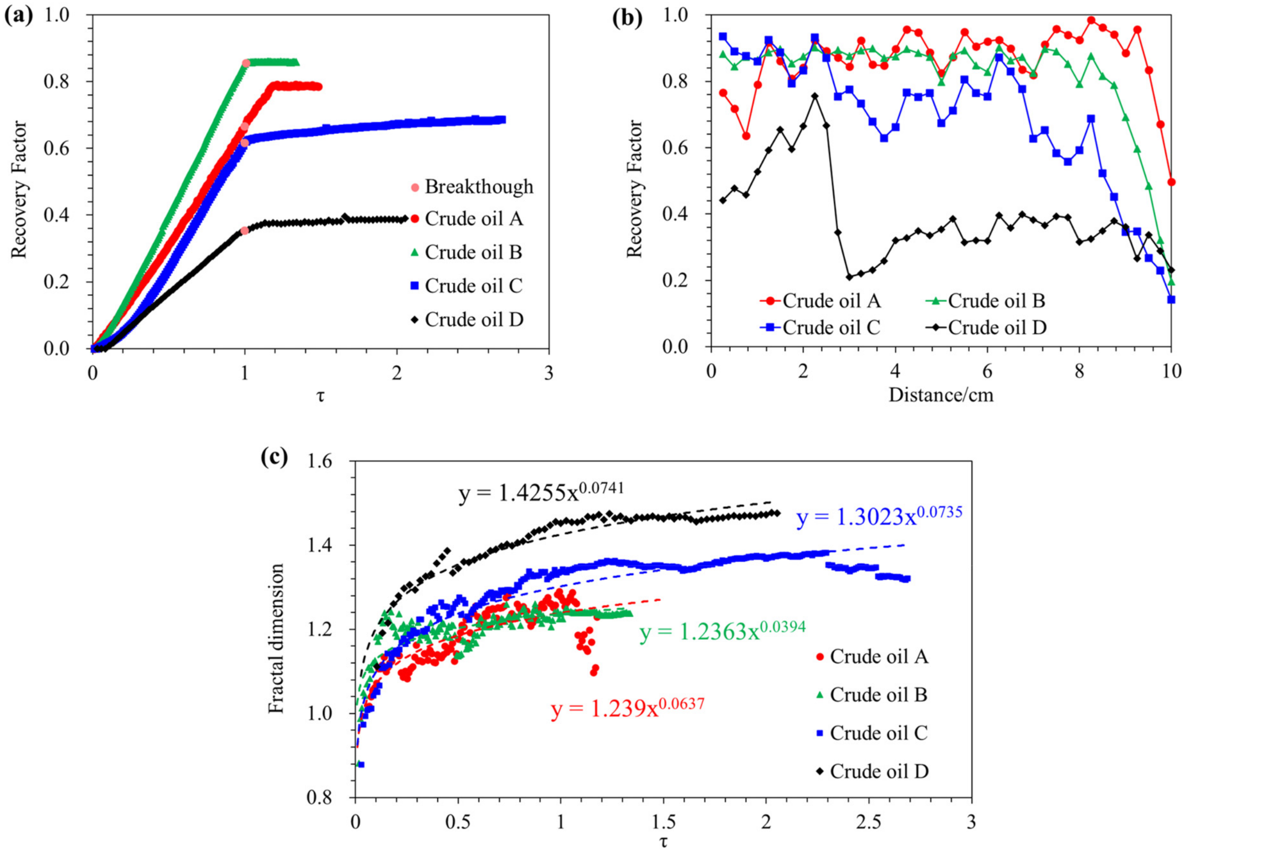

FIG. 5. Characteristic curves of crude oil displacement under different conditions. (a) Relationship between the recovery factor and dimensionless time. (b) Variation of the recovery factor with distance from the inlet at the end of the experiment. (c) Relationship between the fractal dimension of the displacement front and dimensionless time.

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A–D are 26, 39, 65, and 80 μm, respectively. When $Ca_{micro} < 10^{-4.09}$, the frequency of two-phase flow in small pores increases and the interface breaks through these pores more readily [Figs. 7(a)–7(d)]. Although small-pore contributions grow under this condition, large pores still control the overall recovery at the end of displacement. For $Ca_{micro} > 10^{-4.09}$, considerable residual oil remains in large pores, which markedly lowers the total recovery factor [Figs. 7(e) and 7(f)].

2. Selective of fluid sweeping paths under different Ca_{micro}

Sweep sequence diagrams under different Ca_{micro} show that the injected fluid always enters the model from the center of the inlet side. When $Ca_{micro} = 10^{-4.33}$, the two-phase front keeps expanding after breakthrough, sweeping additional regions. In contrast, for $Ca_{micro} > 10^{-4.09}$, the displacement virtually ceases once the front reaches the outlet. Under all conditions, the main swept trajectory is almost horizontal, directly linking the inlet and outlet and thus minimizing flow resistance [Figs. 8(a) and 8(h)]. Minimum and maximum resistance paths were defined to analyze the selectivity of the swept paths in the flow process [Eqs. (11)–(13)]

$$\text{Swept path(SP)} = \min \sum_{inlet \to outlet} \text{moment}, \tag{11}$$

$$\text{Maximum resistance path(MaxRP)} = \min \sum_{inlet \to outlet} \text{radius}, \tag{12}$$

$$\text{Minimum resistance path(MinRP)} = \max \sum_{inlet \to outlet} \text{radius}. \tag{13}$$

For $Ca_{micro} = 10^{-3.96}$, $10^{-4.21}$, the swept path almost overlaps the least-resistance trajectory, indicating high flow stability. By contrast, sweeps at $Ca_{micro} = 10^{-3.70}$, $10^{-4.33}$ show noticeable path randomness. Normal distribution fitting results of the pore diameter distribution along the swept paths and the paths of maximum and minimum resistance under different Ca_{micro} values are shown in Fig. 8(i). The average pore radius corresponding to the maximum resistance path is determined to be 84.5 μm, the minimum resistance path is 114 μm, and the pore radius corresponding to different Ca_{micro} are 106.7, 112, 113.8, and 102.6 μm, respectively. It is evident that although two-phase flow with extreme Ca_{micro} values shows strong randomness, the swept paths still generally follow the principle of minimizing resistance. Notably, even though the swept path of two-phase flow with $Ca_{micro} < 10^{-4.09}$ deviates considerably, it still tends to find paths with larger pore diameters. Only two-phase flow with $Ca_{micro} > 10^{-4.09}$ demonstrates strong randomness [Fig. 8(j)].

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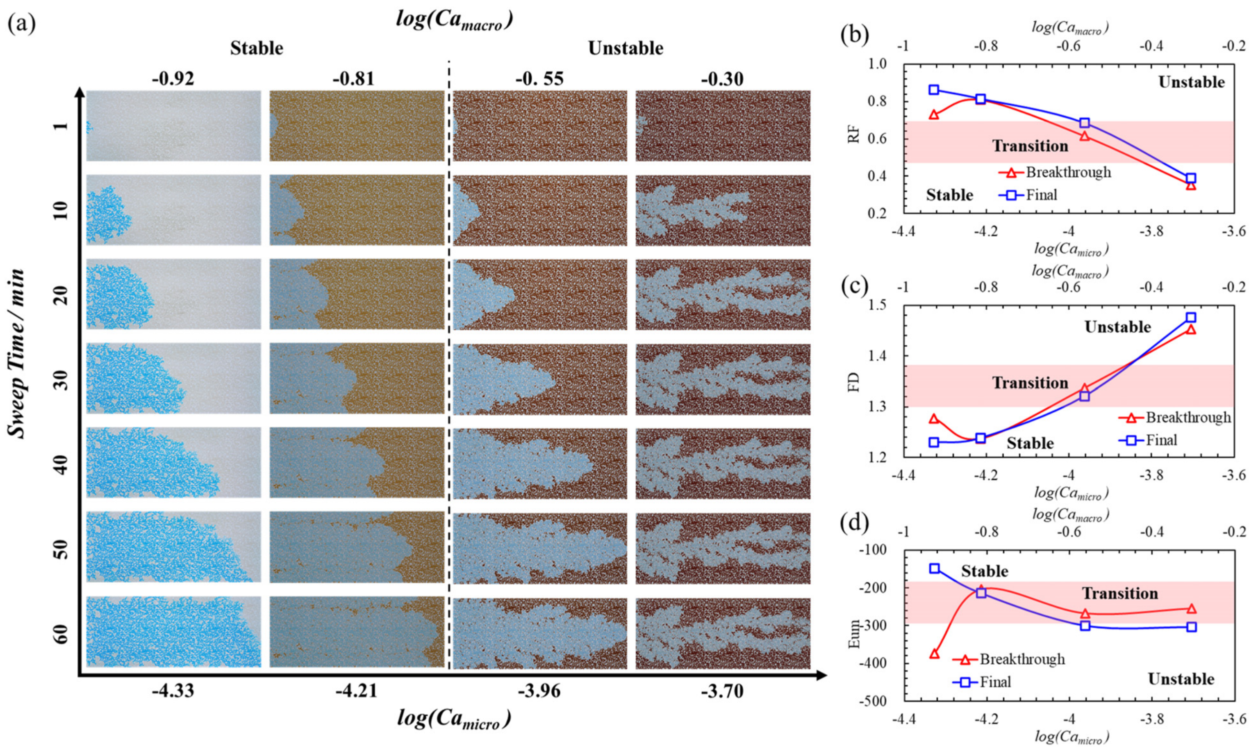

FIG. 6. Displacement diagram in *Ca-Time* space. (a) Displacement pattern and properties of interface in microfluidics with different Ca_{micro} (left to right: $Ca_{micro} = 10^{-4.33}$, $10^{-4.21}$, $10^{-3.96,}$ and $10^{-3.70}$). (b) Recovery factor, (c) Fractal dimension, (d) modified Euler number at different stages of microfluidics with different Ca_{micro}.

3. Pore-scale instability mechanisms controlled by competition between viscous and capillary forces

a. Capillary-dominated region (Flow with $Ca_{micro} = 10^{-4.33}$*).* At this stage, Ca_{micro} is relatively small, so capillary pressure exceeds viscous stress. The injected fluid must overcome the capillary-entry pressure of each pore to advance. Theoretically, the interface selects the pore throat with the lowest "lowest threshold pressure" large pore throat (typically the largest one), where it ruptures via a Haines jump or Snap-off event [Figs. 9(a) and 9(c)]. The resulting local pressure drop rebounds rapidly, driving the front along the minimum-resistance pathway and leaving fingerlike water protrusions [Fig. 9(e)] together with isolated oil islands trapped by the Jamin effect [Fig. 9(f)].[57] The observed sweep pattern, therefore, reflects this pressure-threshold selection. Nevertheless, Sec. IV C 2 reveals occasional deviations from the lowest threshold pressure rule. This research analyzes that the pressure disturbance caused by the rapid rebound of the local pressure drop during the Haines jump/Snap-off causes the interface to break through the adjacent smaller radius pores [Fig. 9(d)], that is, capillary burst. Subsequent flow progressively heals these disturbances, as evidenced by the disappearance of instabilities in Figs. 6(b) and 6(d) and by the higher recovery factor

$$\Delta P_{c,i} = \frac{2\sigma\cos\theta}{r_i}. \tag{14}$$

b. Viscous-dominated region (flow with $Ca_{micro} = 10^{-3.70}$*).* When Ca_{micro} is relatively high, viscous stress at the displacement front exceeds capillary pressure. This stress imbalance triggers Saffman–Taylor instability, producing viscous fingers [Figs. 10(a) and 10(b)]. Independent of pore-size distribution or local pressure rebounds, the non-wetting phase advances along the steepest pressure gradient, generating long, regular yet highly branched fingers. Flow between fingers is negligible, leaving extensive unswept zones [Fig. 10(c)]. Swept path analysis (Fig. 8) confirms the resulting interface disorder, characterized by scattered residual oil, a low recovery factor, and a large fractal dimension.

c. Synergism region (flow with $Ca_{micro} = 10^{-3.96}$, $10^{-4.21}$*).* At this stage, the oil–water front achieves a dynamic balance between Saffman–Taylor instability[58,59] and capillary bursts. Interface selection now strictly follows the principle of lowest threshold pressure, and the swept path completely overlaps the minimum resistance path-an observation termed the "sequential breakthrough effect" (Fig. 11). This finding is consistent with the pore-scale experiments of Lenormand[39] and Chen *et al.*,[12] their research also indicates the existence of these three regions. However, this study found some different conclusions, such as capillary burst and sequential breakthrough effect, and to some extent, these findings are supported by data, instead of more qualitative human judgment. At the same time, this research provides a quantitative pore-scale perspective to those studies. Viscous fingering reduces recovery far more than capillary fingering, whose influence weakens as displacement proceeds. Consequently, the recovery factor for the lowest-viscosity oil (crude A) continues to rise even after breakthrough [Fig. 5(a)].

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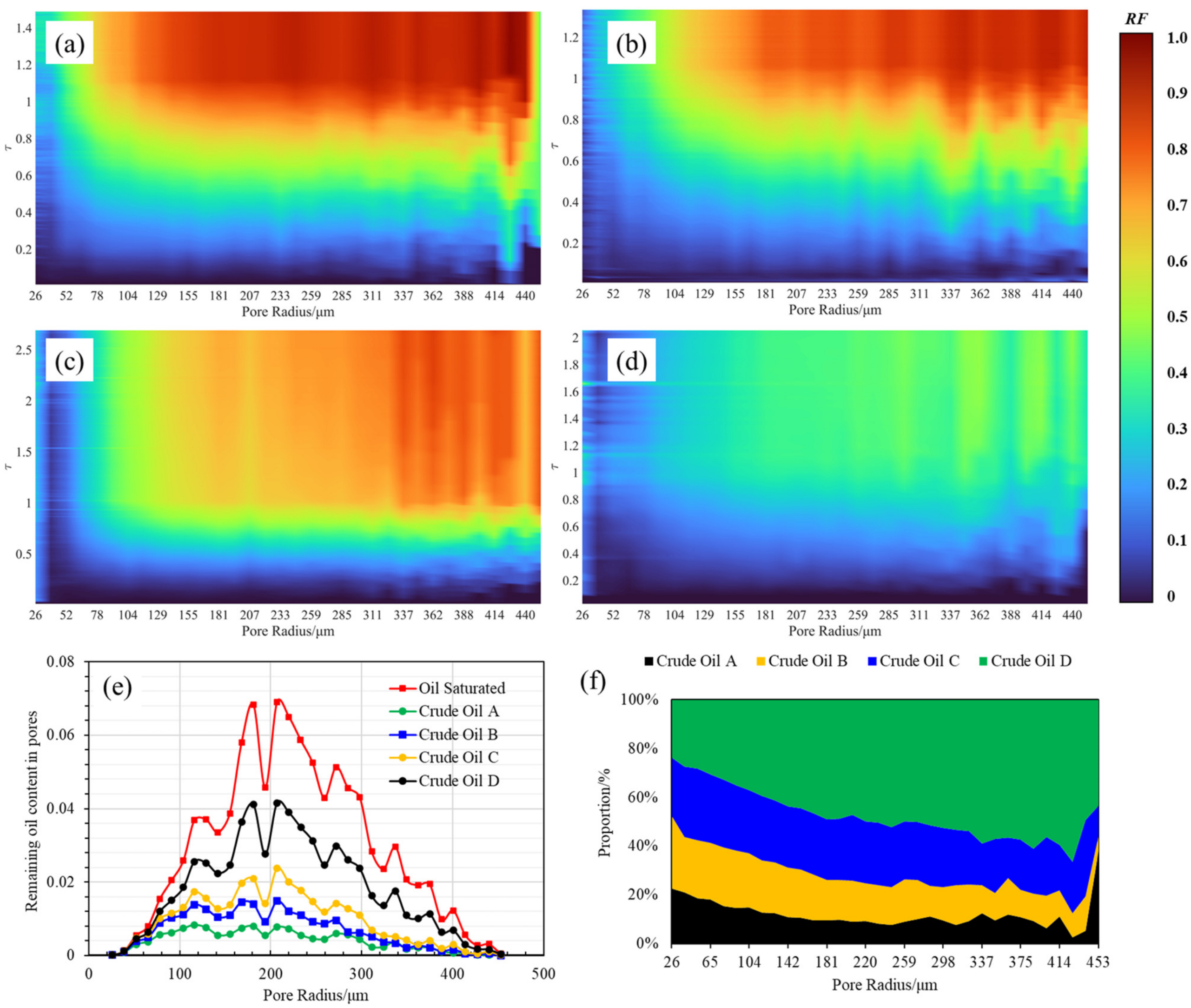

FIG. 7. Recovery factor of pore throat during different crude oil displacement processes. (a)–(d) Heat map of different recovery factors of pore throat with dimensionless time in crude oil A, B, C, and D. (e) Remaining oil content in pores at the final moment of displacement of different types of crude oil. (f) Percentage stacked area chart of the remaining oil content in pores at the final moment of different types of crude oil.

D. Transition process of two-phase flow dynamics mechanism with different Ca_{micro}

The average contact angles of the four crude oils during the oil–water two-phase flow process initially decrease and then stabilize [Fig. 12(a)], eventually settling at 64°. Early in displacement, high flow rates give viscous forces the upper hand, producing dynamic wetting, larger contact angles, and occasional oil–wet behavior. As the system stabilizes, viscous control weakens and the contact angle converges to a constant value in the water-wet range. To quantify the impact of dynamic wetting, this study introduces the dynamic wetting degree, defined as the difference between the initial contact angle and the equilibrium contact angle. When Ca_{micro} increases from $10^{-4.33}$ to $10^{-3.70}$, the dynamic wetting degree increases from 29° to 46°, showing that higher Ca_{micro} amplifies dynamic wetting [Fig. 12(a)]. During the displacement process, the contact angle follows a normal distribution pattern. As the displacement progresses, some areas still exhibit dynamic wetting due to water flow, while most areas are close to equilibrium, which is reflected in the gradual increase in the standard deviation of the contact angle [Figs. 12(c) and 12(f)].

Capillary pressure likewise remains normally distributed, but both its mean rise steadily during displacement [Fig. 12(b)]. The interface is continuously modulated by the dynamic balance between viscous and capillary forces. As the system evolves toward a residual-oil configuration, viscous effects fade and capillary control strengthens [Figs. 12(g) and 12(j)]. Ultimately, the Jamin effect traps oil within pores.

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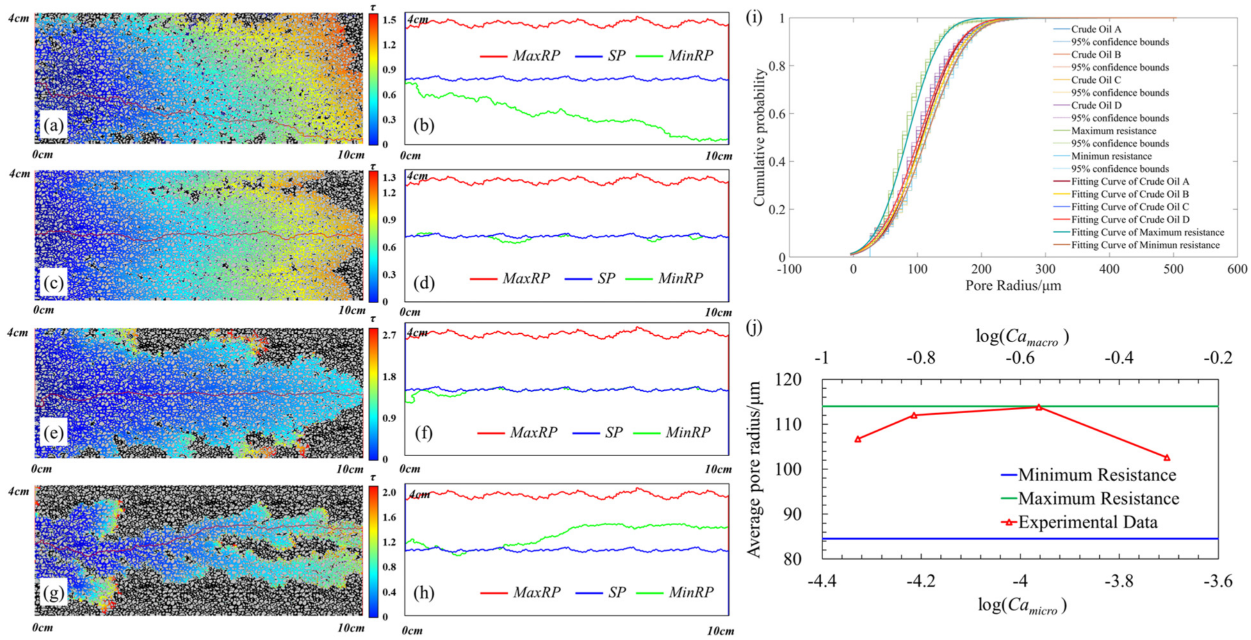

FIG. 8. Analysis of swept paths under different Ca_{micro} conditions. (a), (c), (e), and (g) The sweep sequence diagram of crude oil A, B, C, and D. (b), (d), (f), and (h) Comparison between the swept path and the maximum and minimum resistance paths. (i) A cumulative probability distribution diagram representing the size of pores traversed by different Ca_{micro} swept paths. (j) The relationship between the average pore radius traversed by the swept path and Ca_{micro}.

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V. CONCLUSION AND DISCUSSION

A. Conclusion

This study systematically analyzed the two-phase flow characteristics in displacement processes with varying Ca_{micro}, focusing on stability, sweep path, pore throat mobilization, and the mechanical mechanisms governing oil–water flow. The key conclusions are as follows:

(1) As Ca_{micro} decreases, the recovery factor increases, and the displacement transitions from stable to unstable, with a critical

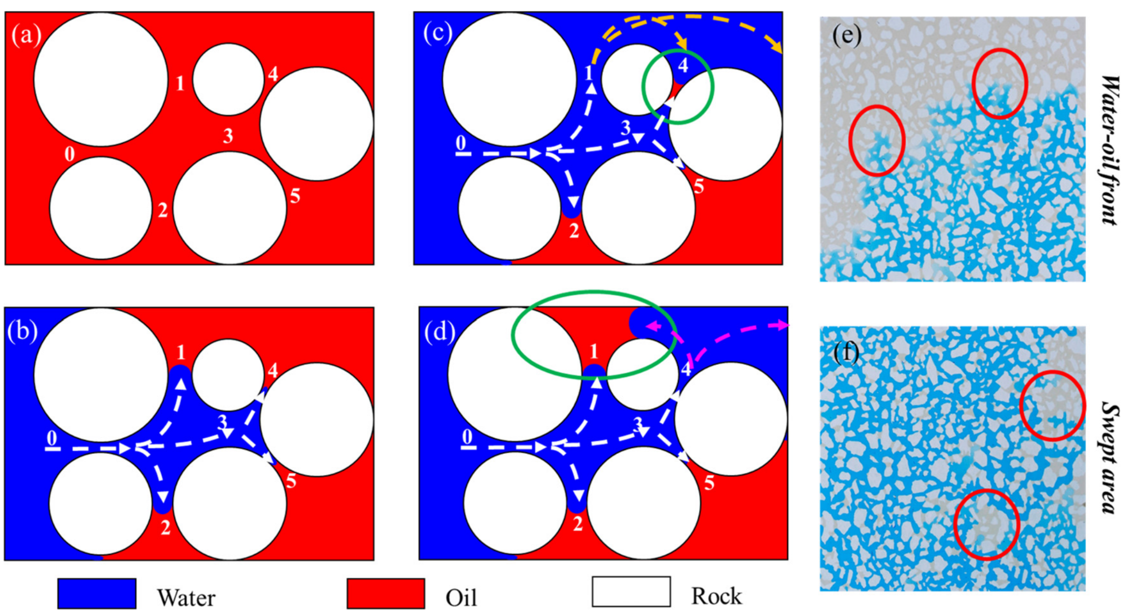

FIG. 9. Interface selectivity analysis in capillary capillary-dominated region. (a)–(d) Mechanism of remaining oil formation under capillary force dominance. (e) The oil–water front under capillary force exhibits finger-like protrusions. (f) Remaining oil in the swept area is dominated by capillary force.

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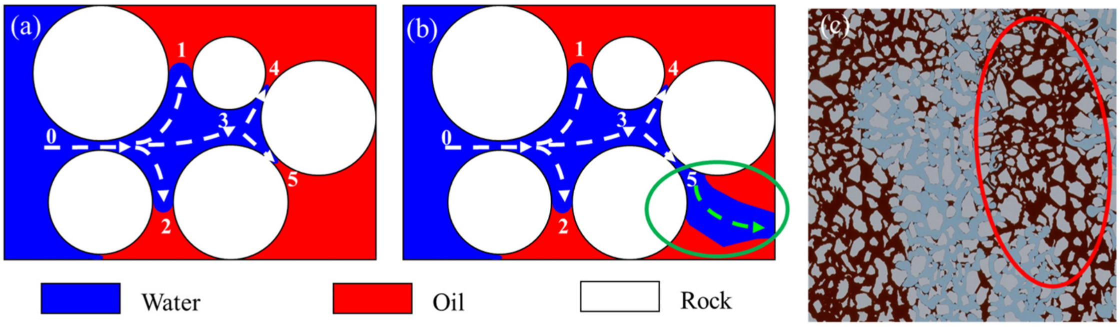

FIG. 10. Interface selectivity analysis in viscous-dominated region. (a) and (b) Mechanism of remaining oil formation under viscous force dominance. (c) Remaining oil in the swept area is dominated by viscous forces.

Ca_{micro} of $10^{-4.09}$. Flows with $Ca_{micro} > 10^{-4.09}$ exhibit greater instability, a faster instability rate, and a smaller swept area. In contrast, flows with $Ca_{micro} < 10^{-4.09}$ sweep a larger area but still leave significant residual oil in the swept area.

(2) Under the combined influence of viscous force and capillary force, two-phase flow in porous media can be classified into capillary-dominated, viscous-dominated, and synergism regions. The rapid rebound of local pressure caused by capillary force dominance leads to capillary burst, which makes the selection of fluid interface deviate from the lowest threshold pressure rule. The Saffman–Taylor instability under the influence of viscous force causes fluid fingering. The sequential breakthrough effect, governed by the combined influence of those factors, enhances the efficiency of medium Ca_{micro} flows in the swept area. However, as displacement progresses, capillary bursts gradually disappear.

(3) Throughout displacement, viscous and capillary forces shape the fluid interface and govern the ultimate recovery factor. As displacement continues, the average contact angle decreases, and viscous-driven dynamic wetting weakens. Oil and water become increasingly dispersed, and the capillary-controlled Jamin effect ultimately fixes the spatial distribution of residual oil.

B. Limitations and future works

Although this study provides a pore-scale method for quantitative analysis of porous media flow, there are still some limitations:

(1) The algorithm proposed in this paper aims to offer an initial approach for pore-scale analysis, including the calculation of contact angles and capillary pressure, the analysis of swept paths, and the calculation of recovery factor. It does not take into account the image processing in very complex experimental

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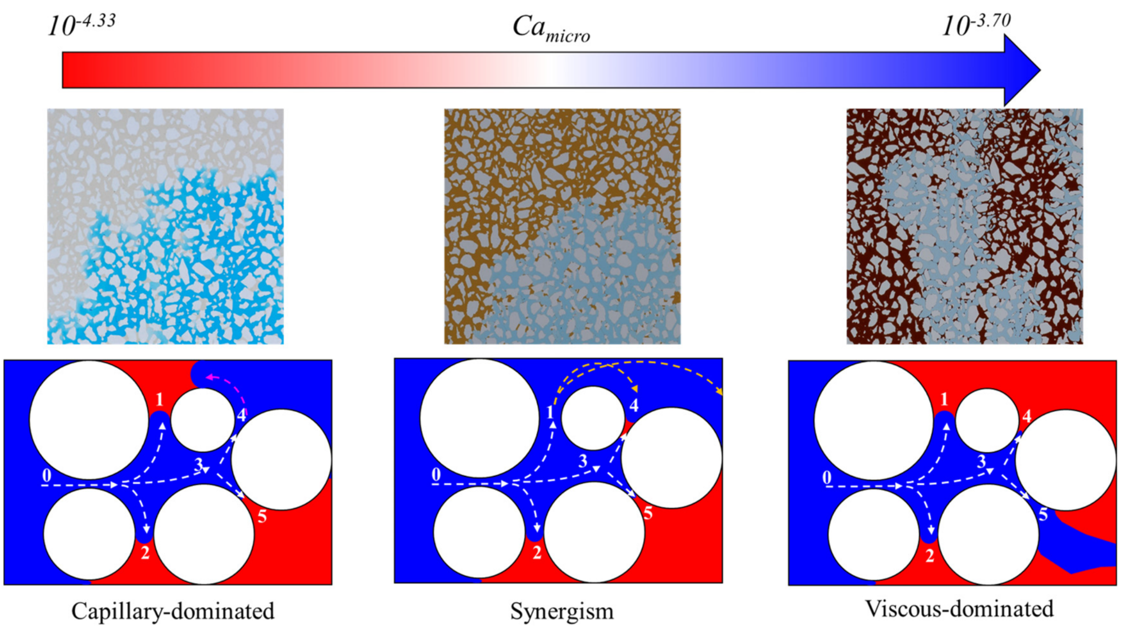

FIG. 11. Flow patterns under different Ca_{micro}.

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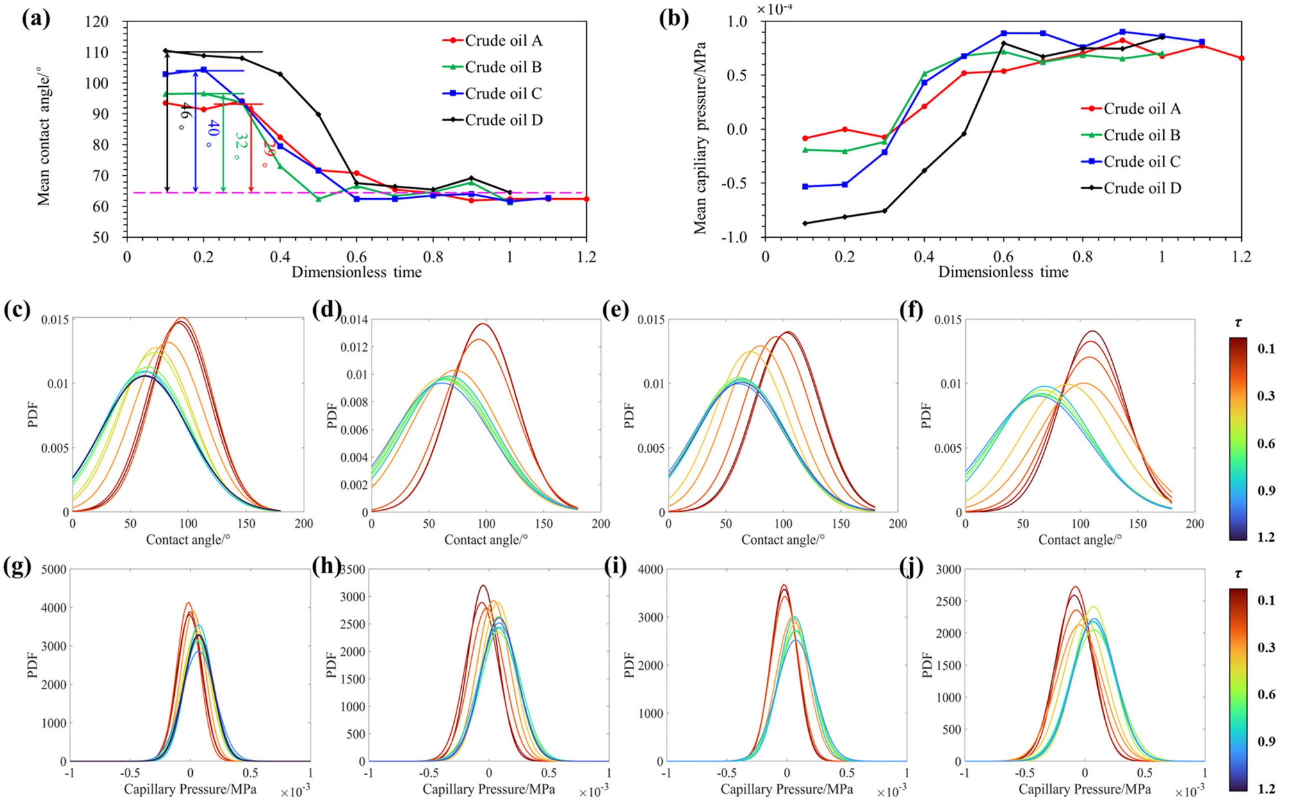

FIG. 12. Mechanical mechanism transitions in the two-phase flow process. (a) Variation of average contact angle with displacement time. (b) Variation of average capillary pressure with displacement time. (c)–(f) Normal distribution fitting results of contact angle measurements within the model during four different crude oil experiments. (g)–(j) Normal distribution fitting results of capillary pressure measurements within the model during four different crude oil experiments.

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environments. Although the algorithm itself may have some issues, the data obtained from the results of this paper are still in line with the general understanding. Subsequent scholars can improve upon this basis to make it adaptable to various complex conditions and achieve more accurate results.

(2) Due to various limitations, some of the analyses in this research, such as the classification of stable and unstable conditions and the analysis of pore-scale mechanisms, are still rather qualitative. In the future, we will consider combining more professional theoretical calculations (such as correlating the dynamic contact angle with the local velocity of flow or contact line and other contact angle measurement methods) and numerical simulation techniques to analyze the mechanisms and critical characterization parameters of pore-scale unstable flow.

(3) This study was conducted flow experiments under conditions of varying viscosity and interfacial tension. In the future, we will continue to consider experiments with more viscosities and interfacial tensions, as well as different flow rates, to expand the universality of this type of research.

SUPPLEMENTARY MATERIAL

See the supplementary material for contents not mentioned in the text, including the detailed principles of image processing methods and some additional pictures involved in this article.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jinxin Cao: Conceptualization (lead); Investigation (lead); Methodology (lead); Software (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Yiqiang Li:** Funding acquisition (lead); Project administration (equal); Resources (equal); Supervision (equal). **Yaqian Zhang:** Data curation (equal); Validation (equal); Visualization (equal); Writing – review & editing (equal). **Xuechen Tang:** Data curation (lead); Formal analysis (equal); Investigation (equal); Writing – review & editing (equal). **Zheyu Liu:** Conceptualization (equal); Investigation (equal); Resources (equal); Supervision (equal); Writing – review & editing (equal). **Qihang Li:** Validation (equal); Writing – review & editing (equal). **Tao Song:** Investigation (equal); Visualization (equal); Writing – review & editing (equal).

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Yuling Zhang: Data curation (equal); Investigation (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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